POWER AND FILTERING

VBUS
VBAT
DMG3415UFY
Q3
D4
PMEG2020AEA
R4 10K
C1 10uF/16V
R3 10K
VHI
EN
U2
IN
EN
GND
OUT
P4
RT9080-3.3
C4 10uF/16V
C5 10uF/16V
+3V3
+3V3
GND

QSPI_CS
RESET RED LED

1K R11
1N5819 D3
USBBOOT
TP3
RKB2
SW2
RESET
RKB2
SW1
D13
10K R7
RED
L0
GND

USB AND BATTERY CHARGING

10K = 100mA
5.0K = 200mA
2.0K = 500mA
1.0K = 1000mA

VBUS
ORANGE
CHG0
R2 10K
U3
MCP73831/2
LIPO Charger
VDD
STAT
VBAT
PROG
VSS
VDD: 3.75-6V
Temp: -40-85°C
MCP73831T-2ACI/OT
VBAT
VBAT
1.5K R8
2K R9
C20 1uF/16V
CON_JST_PH_2PIN_MT_BATT
CON_JST_PH_2PIN_MT_BATT
X1A
X1B
CON_JST_PH_2PIN_MT_BATT
GND
GND
GND

NEOPIXEL

NEOPIXEL_PWR
WS2812B
VDD
NEOPIXEL
DI
GND
DO
GND

RP2040 BYPASS CAPS

1.2V 0.1uF C18
1.2V 1uF/25V C8
1.2V 0.1uF C6
1.2V 0.1uF C7
+3V3 1uF/25V C17
+3V3 1uF/25V C10
+3V3 0.1uF C9
+3V3 1uF/25V C11
+3V3 0.1uF C12
+3V3 0.1uF C13
+3V3 0.1uF C14
+3V3 0.1uF C15
+3V3 1uF/25V C16
GND

USB TYPE C

VBUS
D+
D-
22 ohm R10
22 ohm R9
5.1K R13
5.1K R12
VBUS
USB_D+
USB_D-
CC1
CC2
SBU1
SBU2
GND
USB Type C
X6
JP1 SIGNAL ANALYZE
D+1
D-1
USB C
GND
GND
GND

QSPI STORAGE

+3V3
10K R5
U1
QSPI_SCK 6
QSPI_DATA[0] 5
QSPI_DATA[1] 2
QSPI_CS 1
QSPI_DATA[2] 3
QSPI_DATA[3] 7
SCK
MOSI
MISO
SSEL
WP#/IO2
HOLD#/IO3
VCC 8
VSS
8MB QSPI
+3V3
GND

BREAKOUTS

+3V3
GND
RESET
A0
A1
A2
A3
D24
D25
SCK
MOSI
MISO
RX
TX
D4
JP1
VBAT
VBUS
JP3
EN
D13
D12
D11
D10
D9
D6
D5
SCL
SDA

+3V3
10K R1
RESET
/RUN
IOVDD
USB_IOVDD
VREG_IOVDD
ADC_IOVDD
DVDD
VREG_VOUT
1.2V
GND
C2 22pF
Y1 12MHz
22pF C3
12pF
R6 1K
TESTEN
XIN
XOUT
GPIO24
GPIO25
GPIO26/AD0
GPIO27/AD1
GPIO28/AD2
GPIO29/AD3
QSPI_SCK
QSPI_CS
QSPI_DATA[0]
QSPI_DATA[1]
QSPI_DATA[2]
QSPI_DATA[3]
QSPI_SCLK
QSPI_CS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
P$1
GND
RP2040
RP2040_QFN56
IC1
GPIO0 2 TX
GPIO1 3 RX
GPIO2 4 SDA
GPIO3 5 SCL
GPIO4 6 D4
GPIO5 7 D5
GPIO6 8 D6
GPIO7 9 USBBOOT
GPIO8 11 MISO
GPIO9 12 D9
GPIO10 13 D10
GPIO11 14 D11
GPIO12 15 D12
GPIO13 16 D13
GPIO14 17 SCK
GPIO15 18 MOSI
GPIO16 27 EPD_BUSY
GPIO17 28 EPD_RST
GPIO18 29 EPD_DC
GPIO19 30 EPD_CS
GPIO20 31 NEOPIX_PWR
GPIO21 32 NEOPIXEL
GPIO22 34 EPD_SCK
GPIO23 35 EPD_MOSI
USB_D- 46 D-
USB_D+ 47 D+
SWCLK 24 SWCLK
SWDIO 25 SWDIO

+3V3
GND
MT2 MT
MT1 MT
GND
V+
SDA
SCL
SDA
SCL

+3V3
L1 10uH
GDR
RESE
Q5
D5 1N5819
VGH
1uF/25V C25
D1 1N5819
10uF/16V C27
D2 1N5819
VGL
1uF/25V C26
R14 470ohm
GND
GND
GND
GND

SPRT1 MT
SPRT2 MT
GND
EPD0231
2.13" 104x212
elnk Display
GDR 2
RESE 3
TSCL 6
TSDA 7
EPD_BUSY 9
EPD_RST 10
EPD_DC 11
EPD_CS 12
EPD_SCK 13
EPD_MOSI 14
BUSY
RES#
D/C#
CS#
SCK
SDIO
VGL/NC 4
VSH2 5
BSI 8
VDDIO 15
VCI 16
VSS 17
VDD 18
VPP 19
VSH1 20
VGH 21
VSL 22
VGL 23
VCOM 24
Temp: 0--50°C
VCI: 2.4-3.3V
VDD: 1.7-1.9V
10uF/16V C28
C19
+3V3
10uF/16V C16
1uF/25V C29
VGH
VGL
1uF/25V C24
GND

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${PROJECTNAME}
${CURRENT_DATE}
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Sheet: /
File: working.kicad_sch
Adafruit Industries
Title:
Size: User
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.6-7.0.6-ubuntu22.04.1
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